$$K_h = \frac{[CH_3COOH][OH^-]}{[CH_3COO^-][H_2O]}$$

$$K_h = \frac{[CH_3COOH][OH^-]}{[CH_3COO^-]} \quad .....(1)$$

$$CH_3COOH\ (aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} \quad .....(2)$$

(1) × (2)

$\Rightarrow K_h.K_a = [H^+][OH^-]$

we know that $[H^+][OH^-] = K_w$

$K_h.K_a = K_w$

K_h value in terms of degree of hydrolysis (h) and the concentration of salt (C) for the equilibrium can be obtained as in the case of ostwald's dilution law. $K_h = h^2C$. and

i.e $[OH^-] = \sqrt{K_h.C}$

pH of salt solution in terms of K_a and the concentration of the electrolyte.

pH + pOH = 14

pH = 14 - p OH = 14 - {-log $[OH^-]$}

= 14 + log $[OH^-]$

$\therefore$ pH = 14 + log $(K_hC)^{1/2}$

$$pH = 14 + \log\left(\frac{K_wC}{K_a}\right)^{1/2}$$

$$pH = 14 + \left(\frac{1}{2}\log K_w + \frac{1}{2}\log C - \frac{1}{2}\log K_a\right)$$

$$pH = 14 - 7 + \frac{1}{2}\log C + \frac{1}{2}pK_a$$

$$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C.$$

[$\because K_w = 10^{-14}$.

$\frac{1}{2}\log K_w = \frac{1}{2} \times \log 10^{-14} = \frac{-14}{2}(1) = -7.$

$-\log K_a = pK_a$]

8.8.3 Hydrolysis of salt of strong acid and weak base (Cationic Hydrolysis)

Let us consider the reactions between a strong acid, HCl, and a weak base, NH_4OH, to produce a salt, NH_4Cl, and water

$$HCl\ (aq) + NH_4OH\ (aq) \rightleftharpoons NH_4Cl(aq) + H_2O(l)$$

$$NH_4Cl(aq) \rightarrow NH_4^+ + Cl^-(aq)$$

NH_4^+ is a strong conjugate acid of the weak base NH_4OH and it has a tendency to react with OH^- from water to produce unionised NH_4OH shown below.

$$NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_4OH\ (aq) + H^+(aq)$$

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